

Experiment 2 – MDP

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COURSE CODE:MLA0305

COURSE NAME:REINFORCEMENT LEARNING

SOLT:B

2.Implementation of Reinforcement Learning Framework and Markov Decision Process (MDP)

Aim:

To implement a basic **Reinforcement Learning (RL) framework** using a **Markov Decision Process (MDP)** and understand the interaction between an agent and its environment through states, actions, rewards, and transitions.

Procedure:

1. Define the **environment** and represent it using a set of states.
2. Define the possible **actions** that the agent can perform in each state.
3. Assign **rewards** for different state transitions.
4. Define the **transition process** between states based on the selected actions.
5. Initialize the agent at a starting state.
6. Allow the agent to select an action and interact with the environment.
7. Calculate the reward obtained after each action.
8. Move the agent to the next state according to the transition rules.
9. Continue the process until the agent reaches the goal state or the episode ends.
10. Display the states, actions, rewards, and final result to demonstrate the MDP-based RL framework.

R-CODE:

```
# Markov Decision Process
```

```
grid_size <- 4
```

```
start <- c(1,1)
```

```
goal <- c(4,4)
```

```
state <- start
```

```
# Function to draw coloured grid
```

```
drawGrid <- function(state, step){
```

```
  grid.newpage()
```

```
  grid.text(
```

```
    paste("Markov Decision Process - Step", step),
```

```
    y=0.97,
```

```
    gp=gpar(fontsize=18,fontface="bold")
```

```
  )
```

```
  for(i in 1:grid_size){
```

```
    for(j in 1:grid_size){
```

```
      fill <- "white"
```

```
      if(i==start[1] && j==start[2])
```

```
        fill <- "skyblue"
```

```
      if(i==goal[1] && j==goal[2])
```

```
        fill <- "lightgreen"
```

```
        if(i==state[1] && j==state[2])
```

```
          fill <- "red"
```

```

x <- (j-0.5)/grid_size
y <- 1-(i-0.5)/grid_size

grid.rect(
  x=x,
  y=y,
  width=1/grid_size,
  height=1/grid_size,
  gp=gpar(fill=fill,col="black",lwd=2)
)

if(i==start[1] && j==start[2])
  grid.text("S",x,y,gp=gpar(fontsize=18,fontface="bold"))

if(i==goal[1] && j==goal[2])
  grid.text("G",x,y,gp=gpar(fontsize=18,fontface="bold"))

if(i==state[1] && j==state[2])
  grid.text("A",x,y,
    gp=gpar(col="white",
      fontsize=18,
      fontface="bold"))
}
}

# Legend
grid.text("Blue = Start  Red = Agent  Green = Goal",
  y=0.03,

```

```

gp=gpar(fontsize=12,fontface="bold"))
}

# Simulation

step <- 0

total_reward <- 0

cat("=====\n")

cat(" MARKOV DECISION PROCESS SIMULATION\n")

cat("=====\n\n")

drawGrid(state, step)

cat("Step :", step,"\n")

cat("Current State :", state,"\n")

cat("Action : Start\n")

cat("Reward : 0\n\n")

Sys.sleep(1)

while(!all(state==goal)){

  current <- state

  if(state[1] < 4){

    state <- c(state[1]+1,state[2])

    action <- "Down"

  }else{

    state <- c(state[1],state[2]+1)

    action <- "Right"

  }

  step <- step + 1

```

```

if(all(state==goal))

    reward <- 100

else

    reward <- -1

total_reward <- total_reward + reward

# Visual Output

drawGrid(state, step)

# Console Summary

cat("-----\n")

cat("Step      :", step, "\n")

cat("Current State :", current, "\n")

cat("Action      :", action, "\n")

cat("Next State   :", state, "\n")

cat("Reward       :", reward, "\n\n")

Sys.sleep(1)

}

cat("=====\n")

cat(" FINAL SUMMARY\n")

cat("=====\n")

cat("Start State  :", start, "\n")

cat("Goal State   :", goal, "\n")

cat("Final State  :", state, "\n")

cat("Total Steps  :", step, "\n")

cat("Total Reward :", total_reward, "\n")

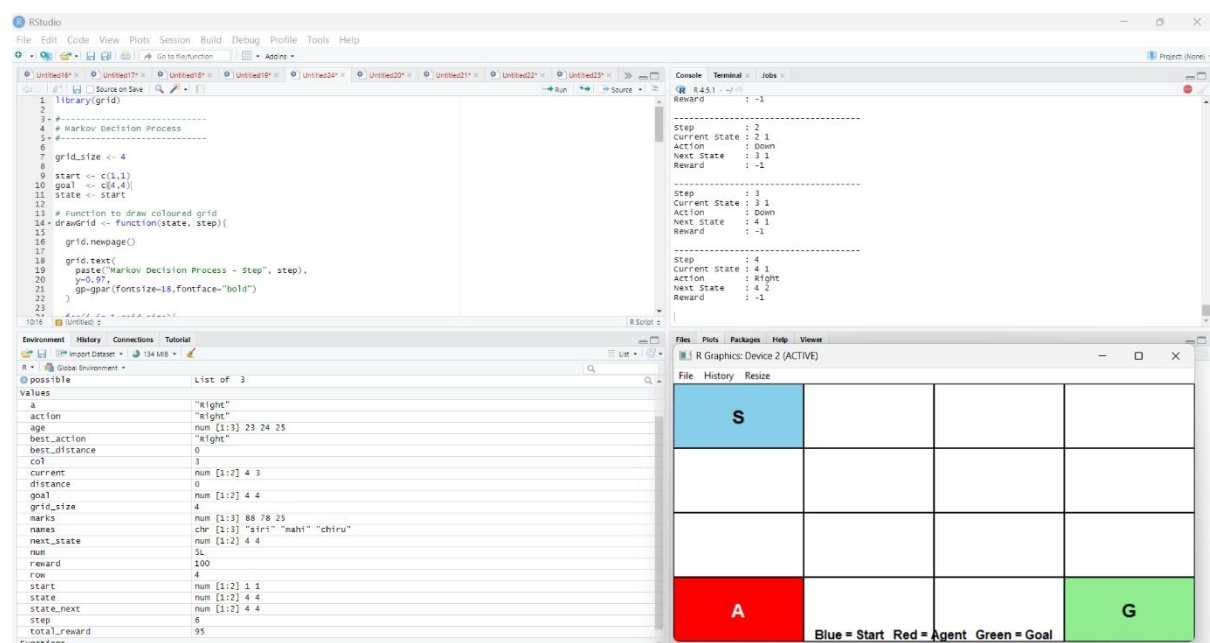
```

```
cat("Goal Reached : YES\n")
```

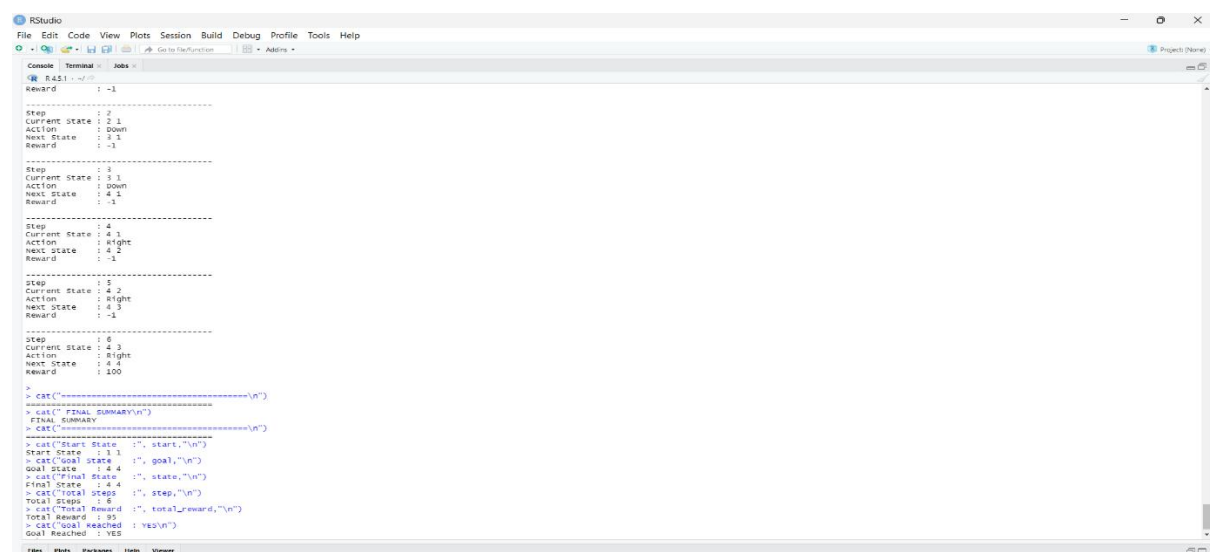
Result:

The **Reinforcement Learning framework and Markov Decision Process (MDP)** were successfully implemented. The agent interacted with the environment by selecting actions, receiving rewards, and transitioning between states. The implementation demonstrates how an RL agent learns to reach a desired goal through **states, actions, rewards, and state transitions**.

Full screenshot of Rstudio:



Summary Result:



Visualization Result:

